

Foundations of Flow-based generative model

[Flow-based generative model]

1.0 Content Block

$x = f^{\text{trans}} - 1(y; c) = \mathbf{y} - c$, where $\mathbf{y} = \mathbf{y}'c + (\mathbf{y}'c)^2 + 1 - c'c$ $\{\text{displaystyle } \mathbf{x} = f^{\text{trans}} - 1(\mathbf{y}; \mathbf{c}) = \mathbf{y} - \mathbf{c} \text{ , where } \mathbf{y} = \mathbf{y}'c + (\mathbf{y}'c)^2 + 1 - c'c \}$

2.0 Content Block

To understand the geometric interpretation of the factor n $\{\text{displaystyle } \sqrt{n}\}$, see the example for the 1-simplex in the diagram at right. The differential volume element at the transformation output ($q \in \Delta_{n-1}$ $\{\text{displaystyle } \mathbf{q} \in \Delta^{n-1}\}$), is the parallelotope, $V = f(U) = F_p E D$ $\{\text{displaystyle } V = f(U) = F_p E D \}$, where F_p $\{\text{displaystyle } \mathbf{F}_p\}$ is the n -by- n $\{\text{displaystyle } n \text{-by-} n \}$ Jacobian of f $\{\text{displaystyle } f\}$ at $p = e(p \sim)$ $\{\text{displaystyle } \mathbf{p} = e(\tilde{\mathbf{p}})\}$. Its volume is:

3.0 Content Block

Since the trace depends only on the diagonal of the Jacobian $\partial z_t f$ $\{\text{displaystyle } \partial_{z_t} f\}$, this allows "free-form" Jacobian. Here, "free-form" means that there is no restriction on the Jacobian's form. It is contrasted with previous discrete models of normalizing flow, where the Jacobian is carefully designed to be only upper- or lower-diagonal, so that the Jacobian can be evaluated efficiently.

4.0 Content Block

$P_Q(q) = P_P(p) R_f(\Delta(p))$, where $p = f^{-1}(q)$ $\{\text{displaystyle } P_Q(\mathbf{q}) = \frac{P_P(\mathbf{p})}{R_f(\Delta(\mathbf{p}))} \text{ , where } \mathbf{p} = f^{-1}(\mathbf{q}) \}$

5.0 Content Block

where the condition that A $\{\text{displaystyle } \mathbf{A}\}$ has $\mathbf{1}$ $\{\text{displaystyle } \mathbf{1}\}$ as an eigenvector ensures invertibility by sidestepping the information loss due to the invariance: $\text{softmax}(\mathbf{x} + \alpha \mathbf{1}) = \text{softmax}(\mathbf{x})$ $\{\text{displaystyle } \operatorname{softmax}(\mathbf{x} + \alpha \mathbf{1}) = \operatorname{softmax}(\mathbf{x})\}$. Note in particular that $A = \lambda I$ $\{\text{displaystyle } \mathbf{A} = \lambda \mathbf{I}\}$ is the only allowed diagonal parametrization, in which case we recover $f_{\text{cal}}(p; \lambda - 1, c)$ $\{\text{displaystyle } f_{\text{cal}}(\mathbf{p}; \lambda - 1, \mathbf{c})\}$, while (for $n > 2$ $\{\text{displaystyle } n > 2\}$) generalization is possible with non-diagonal matrices. The inverse is:

6.0 Content Block

Differential volume ratio For $m < n$ $\{\text{displaystyle } m < n\}$, let $M \subset \mathbb{R}^n$ $\{\text{displaystyle } \mathcal{M} \subset \mathbb{R}^n\}$ be an m $\{\text{displaystyle } m\}$ -dimensional manifold with a smooth, isometric embedding into $\mathbb{R}^n$ $\{\text{displaystyle } \mathbb{R}^n\}$. Let $f: \mathbb{R}^n \rightarrow \mathbb{R}^n$ $\{\text{displaystyle } f: \mathbb{R}^n \rightarrow \mathbb{R}^n\}$ be a smooth flow transform with range restricted to M $\{\text{displaystyle } \mathcal{M}\}$. Let $x \in M$ $\{\text{displaystyle } \mathbf{x} \in \mathcal{M}\}$ be sampled from a distribution with density P_X $\{\text{displaystyle } P_X\}$. Let $y = f(x)$ $\{\text{displaystyle } \mathbf{y} = f(\mathbf{x})\}$, with resultant (pushforward) density P_Y $\{\text{displaystyle } P_Y\}$. Let $U \subset M$ $\{\text{displaystyle } U \subset \mathcal{M}\}$ be a small, convex region containing x $\{\text{displaystyle } \mathbf{x}\}$ and let $V = f(U)$ $\{\text{displaystyle } V = f(U)\}$ be its image, which contains y $\{\text{displaystyle } \mathbf{y}\}$; then by conservation of probability mass:

7.0 Content Block

Derivation of log likelihood Consider z_1 and z_0 . Note that $z_0 = f_{1-1}(z_1)$. By the change of variable formula, the distribution of z_1 is:

8.0 Content Block

In fact, the set of f_{gcal} transformations form a group under function composition. The set of f_{cal} transformations form a subgroup. Also see: Dirichlet calibration, which generalizes f_{gcal} , by not placing any restriction on the matrix, A , so that invertibility is not guaranteed. While Dirichlet calibration is trained as a discriminative model, f_{gcal} can also be trained as part of a generative calibration model.

9.0 Content Block

$p_1(z_1) = p_0(z_0) \left| \det \left(\frac{df_1(z_0)}{dz_0} \right) \right|^{-1}$

10.0 Content Block

$\log(p(x)) = \log(p(z_0)) - \int_0^T \text{Tr} \left[\frac{\partial f}{\partial z} \right] dt$

11.0 Content Block

$\log p_K(z_K) = \log p_0(z_0) - \sum_{i=1}^K \log \left| \det \frac{df_i(z_{i-1})}{dz_{i-1}} \right|$

12.0 Content Block

Since the Jacobian is injective (full rank: m), a local (not necessarily unique) left inverse, say r_x with Jacobian R_x , exists such that $r_x(e_x(x)) = x$ and $R_x T_x = I_m$. In practice we do not need the left inverse function itself, but we do need its Jacobian, for which the above equation does not give a unique solution. We can however enforce a unique solution for the Jacobian by choosing the left inverse as, $r_x: \mathbb{R}^n \rightarrow \mathbb{R}^m$ to

13.0 Content Block

This formula is valid only because the simplex is flat and the Jacobian, E is constant. The more general case for curved manifolds is discussed below, after we present two concrete examples of simplex flow transforms.

14.0 Content Block

Nonlinear Independent Components Estimation (NICE) Let $x, z \in \mathbb{R}^{2n}$ be even-dimensional, and split them in the middle. Then the normalizing flow functions are $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = f_{\theta}(z) = \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} + \begin{bmatrix} 0 \\ m_{\theta}(z_1) \end{bmatrix}$ where m_{θ} is any neural network with weights θ .